

Day 34



The Finally Clause:

What is the Finally Clause?

The finally block is a part of JavaScript's error-handling structure (try → catch → finally). It is used to write code that must run no matter what happens—whether an error occurs or not.

Basic idea

- try → code that may cause an error
- catch → runs only if an error happens
- finally → always runs, after try or catch, no matter what

This is why finally is used for things like closing files, stopping timers, cleaning up memory, resetting variables, etc.

Example: a basic try...catch block without finally.

```
1  try {  
2      let a = 0;  
3      console.log(program); //will create an error  
4      console.log("Program executed successfully.");  
5  } catch (err) {  
6      console.error("This is an error");  
7  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\JavaScript\Day31-40\Day34> node main.js  
This is an error
```

Example: introducing error in the catch block as well as in try block

```
JS main.js > ...
1  try {
2    let a = 0;
3    console.log(program); //will create an error
4    console.log("Program executed successfully.");
5  } catch (err) {
6    console.error("This is an error");
7    console.log(p);
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\JavaScript\Day31-40\Day34> node main.js

This is an error
E:\JavaScript\Day31-40\Day34\main.js:7
console.log(p);
^

So, what if we write some code after the catch block? It will not run:

```
JS main.js > ...
1  try {
2    let a = 0;
3    console.log(program); //will create an error
4    console.log("Program executed successfully.");
5  } catch (err) {
6    console.error("This is an error");
7    console.log(p);
8  }
9  console.log("Program continues...");
```

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PS E:\JavaScript\Day31-40\Day34> node main.js

This is an error
E:\JavaScript\Day31-40\Day34\main.js:7
console.log(p);
^

Then how we make sure that if error occurred in “catch” block then the rest of the code should work?
We can do so by using the “finally” block:

```
JS main.js > ...
1  try {
2    let a = 0;
3    console.log(program); //will create an error
4    console.log("Program executed successfully.");
5  } catch (err) {
6    console.error("This is an error");
7    console.log(p);
8  }
9  finally{
10   console.log("Program continues...");
11 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS E:\JavaScript\Day31-40\Day34> node main.js

This is an error
Program continues...
E:\JavaScript\Day31-40\Day34\main.js:7
console.log(p);
^